

Skills Summary

Right Triangles in Elevation and Depression Problems

Use this reference while completing your worksheet or when studying.

SOH-CAH-TOA, and how to pick one. Label the triangle from the angle you know or want: **opp** is across from it, **adj** is the leg touching it, **hyp** is the line of sight.

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} \quad \cos \theta = \frac{\text{adj}}{\text{hyp}} \quad \tan \theta = \frac{\text{opp}}{\text{adj}}$$

Pick the ratio that uses the side you *have* and the side you *want*; the third side never enters. Both angles are measured from the **horizontal**.

1. Elevation: solving for a side

Looking *up*. The angle sits at the observer, between the horizontal ground and the line of sight.

Ground distance given — it is the *adjacent* leg, so height = distance $\times \tan \theta$.

Slanted length given (rope, string, wire) — it is the *hypotenuse*, so height = length $\times \sin \theta$.

Example: From a point 60 ft from the base of a mast, the angle of elevation to the top is 25° . Find the height.

Step 1. The 60 ft is measured along the ground, so it is adjacent to the angle and the height is opposite it — opposite over adjacent is tangent.

Step 2. $h = 60 \tan 25^\circ = 60(0.466307\dots) = 27.9784\dots \text{ ft} \Rightarrow \mathbf{27.98 \text{ ft}}$

Try it: The angle of elevation to the top of a pole from a point 75 ft away on level ground is 40° . Find the height, to the nearest hundredth.

check: $75 \tan 40^\circ \approx 62.79$

2. Elevation: solving for the angle

Two sides known, angle wanted: write the ratio first, then undo it with the inverse function.

$$\tan \theta = \frac{\text{opp}}{\text{adj}} \quad \Rightarrow \quad \theta = \tan^{-1} \left(\frac{\text{opp}}{\text{adj}} \right)$$

$\tan^{-1}$ (also written arctan) is the *inverse* tangent — it is not $1/\tan$. Build the ratio the right way up, opposite over adjacent, or the angle comes out as its complement. If a problem asks only for the ratio, leave it as a reduced fraction.

Example: A path rises 5 m over a horizontal run of 18 m. Find its angle of elevation.

Step 1. Both legs are known and the angle is the unknown, so form the opposite-over-adjacent ratio and invert it rather than solving for a side.

Step 2. $\tan \theta = \frac{5}{18} = 0.2777\dots$, so $\theta = \tan^{-1}(0.2777\dots) = 15.5241\dots^\circ \Rightarrow \mathbf{15.5^\circ}$

Try it: A ramp rises 7 ft over a run of 24 ft. Find its angle of elevation, to the nearest tenth of a degree.

check: 6.891

3. Depression: move the angle into the triangle

Looking *down*. The angle of depression is measured at the observer from a horizontal line, so it sits *outside* the triangle. That horizontal is parallel to the ground, so by alternate interior angles

$$\text{angle of depression at the top} = \text{angle of elevation at the object}.$$

Move it down to the object first — then it is an ordinary elevation problem. The height is opposite that angle and the ground distance is adjacent, so distance = $\frac{\text{height}}{\tan \theta}$.

Example: From a tower 120 m tall the angle of depression to a car is 25° . How far is the car from the foot of the tower?

Step 1. The angle is outside the triangle, so transfer it to the car as an equal angle of elevation before choosing any ratio.

Step 2. The height is opposite and the distance adjacent, so $\tan 25^\circ = \frac{120}{d}$ and $d = \frac{120}{0.466307\dots} = 257.3408\dots \text{ m} \Rightarrow \mathbf{257.34 \text{ m}}$

Try it: From a cliff 90 m high the angle of depression to a raft is 18° . How far is the raft from the foot of the cliff?

check: 66.927

4. Two triangles sharing a leg

Two objects sighted from one point make two right triangles with the *same* vertical leg. Solve each horizontal distance separately, then combine: subtract for the gap between them, add if they are on opposite sides. **Never subtract the angles.**

Example: From a cliff 40 m high, the angles of depression to two boats in line with the cliff are 45° and 30° . How far apart are they?

Step 1. One cliff, two sight lines, so treat it as two separate right triangles that share the 40 m leg rather than one triangle with a combined angle.

Step 2. $d_{30} = \frac{40}{\tan 30^\circ} = 69.2820\dots$ and $d_{45} = \frac{40}{\tan 45^\circ} = 40$, so the gap is $69.2820\dots - 40 = 29.2820\dots \text{ m} \Rightarrow \mathbf{29.28 \text{ m}}$

Try it: From a cliff 60 m high the angles of depression to two boats in line with the cliff are 50° and 25° . Find the distance between them.

check: 78.82